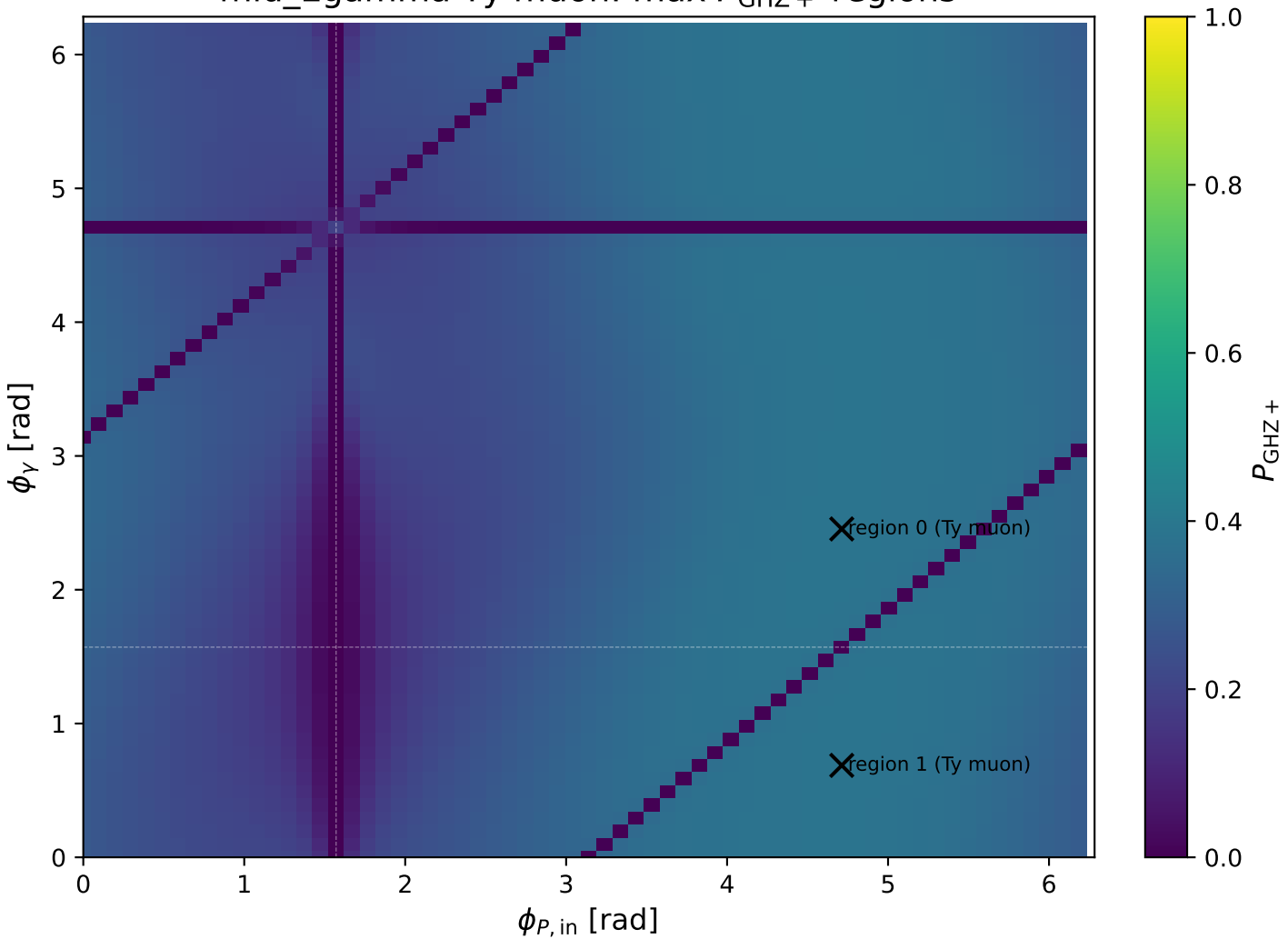
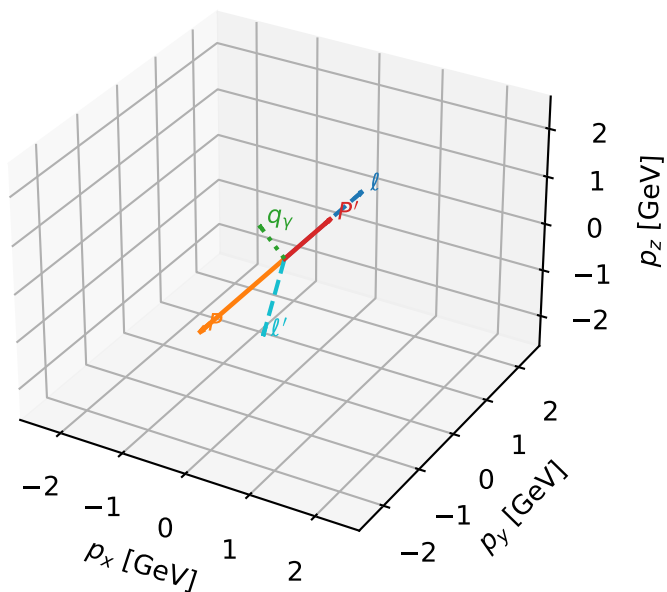


mid_Egamma Ty muon: max $P_{\text{GHZ}+}$ regions



Ty muon characteristic max P_{GHZ} + configuration

3D momenta

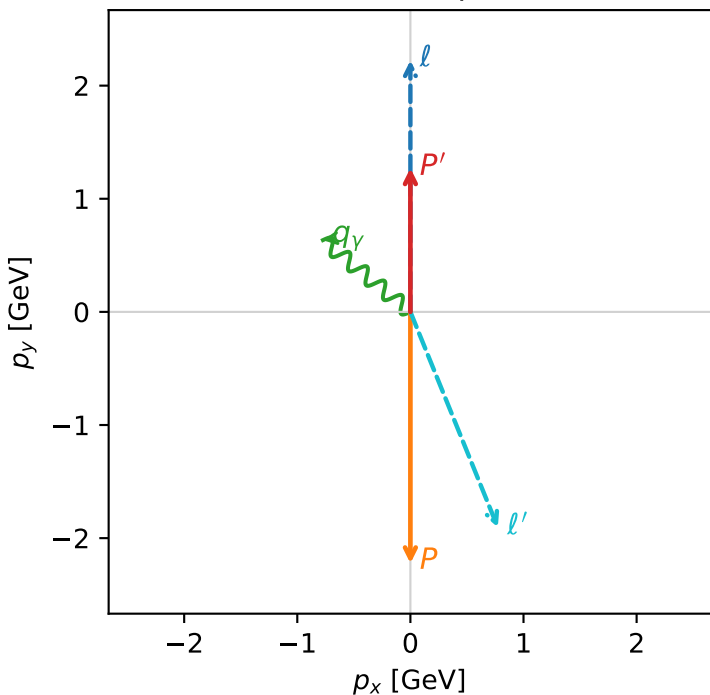


ghz_purity_mid_egamma_ty_lepton_region_0 $P_{\{\text{rm GHZ}\}}=0.384146$
 region: mid_Egamma_Ty_lepton_region_0
 outgoing-state purity: 0.5
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\mu}, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.27096$
 $\theta_{\text{in}}=1.57079$, $\phi_l=1.5708$, $\phi_p=4.71239$
 $E_\gamma=1$, $\phi_\gamma=2.45437$
 $Q^2=17.614$, $x_B=0.91928$, $t=-11.5142$, $W^2=2.4265$, $y=0.929007$
 $\theta(l', \gamma)=151.458$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 0.0000$ $p_y= 2.2231$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -0.0000$ $p_y= -2.2231$ $p_z= 0.0000$
 l' $E= 2.0589$ $p_x= 0.7730$ $p_y= -1.9054$ $p_z= 0.0000$
 P' $E= 1.5796$ $p_x= 0.0000$ $p_y= 1.2710$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.7730$ $p_y= 0.6344$ $p_z= 0.0000$

Transverse plane



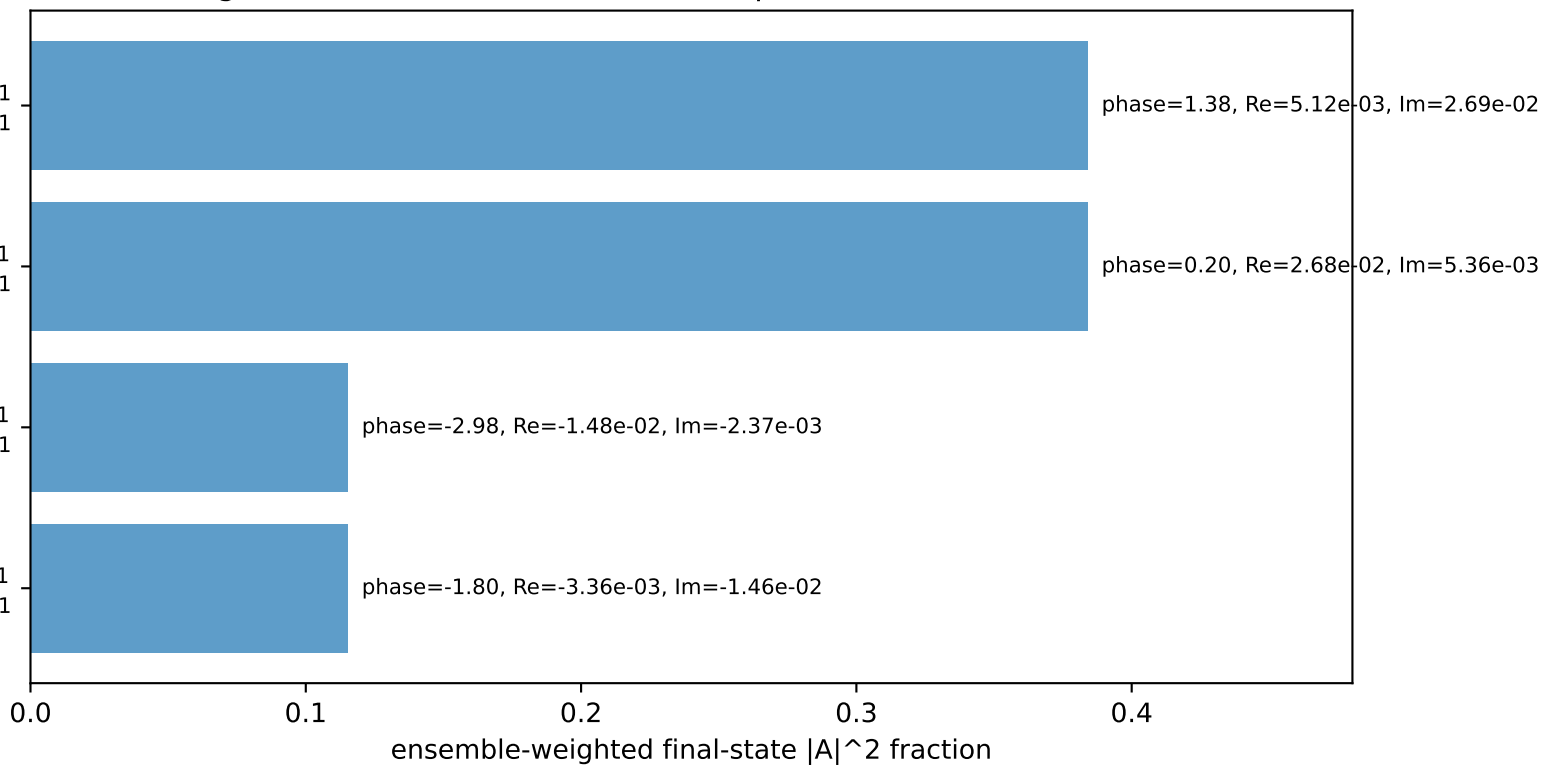
$h_e=+1, h_p=+1, h_\gamma=+1$
 electron Ty, proton h=+1

$h_e=-1, h_p=-1, h_\gamma=-1$
 electron Ty, proton h=-1

$h_e=-1, h_p=-1, h_\gamma=+1$
 electron Ty, proton h=-1

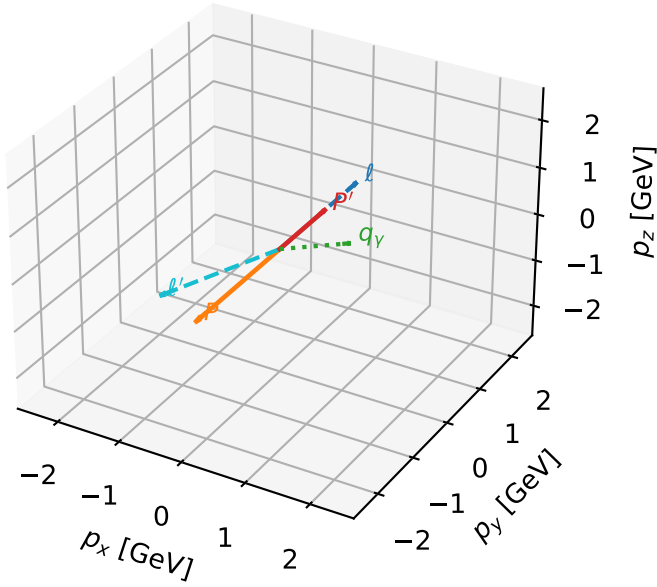
$h_e=+1, h_p=+1, h_\gamma=-1$
 electron Ty, proton h=+1

Leading initial-ensemble/final-state components (N=4, retained=99.9%)



Ty muon characteristic max P_{GHZ} + configuration

3D momenta



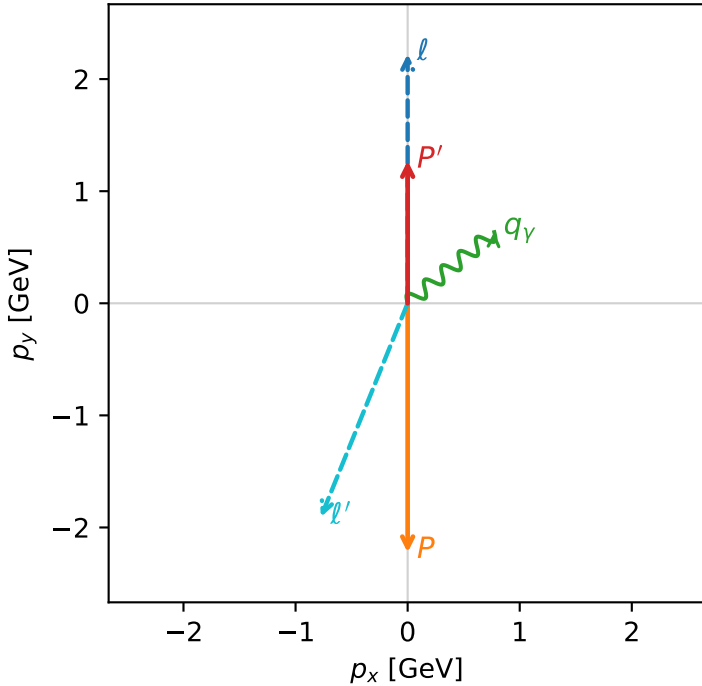
ghz_purity_mid_egamma_ty_lepton_region_1 $P_{\{\text{rm GHZ}\}}=0.384146$
 region: mid_Egamma_Ty_lepton_region_1
 outgoing-state purity: 0.5
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\mu},h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.27096$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=1.5708$, $\phi_P=4.71239$
 $E_{\gamma}=1$, $\phi_{\gamma}=0.687223$
 $Q^2=17.614$, $x_B=0.91928$, $t=-11.5142$, $W^2=2.4265$, $y=0.929007$
 $\theta(\ell',\gamma)=151.458$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 2.2256$ $p_x= 0.0000$ $p_y= 2.2231$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -0.0000$ $p_y= -2.2231$ $p_z= 0.0000$
 ℓ' $E= 2.0589$ $p_x= -0.7730$ $p_y= -1.9054$ $p_z= 0.0000$
 P' $E= 1.5796$ $p_x= 0.0000$ $p_y= 1.2710$ $p_z= 0.0000$
 q_{γ} $E= 1.0000$ $p_x= 0.7730$ $p_y= 0.6344$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=99.9%)

